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| Tool Version : Vivado v.2023.1 (win64) Build 3865809 Sun May  7 15:05:29 MDT 2023
| Date        : Wed Sep  9 13:23:40 2026
| Host        : bedouinoureddine running 64-bit major release (build 9200)
| Command     : report_drc -name drc_1 -file
C:/Vivado_Projects/Harness_Test_System_3/Harness_Test_System/DRC_drc_2.txt
-ruledecks default
| Design      : FPGA_Top
| Device      : xc7a100tcsg324-1
| Speed File  : -1
| Design State : Fully Routed
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```

Report DRC

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- 1. REPORT SUMMARY
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1. REPORT SUMMARY

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Netlist: netlist
Floorplan: impl_1
Design limits: <entire design considered>
Ruledeck: default
Max violations: <unlimited>
Violations found: 6
```

Rule	Severity	Description	Violations
DPIP-1	Warning	Input pipelining	2
DPOP-1	Warning	PREG Output pipelining	2
DPOP-2	Warning	MREG Output pipelining	2

2. REPORT DETAILS

```
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DPIP-1#1 Warning
Input pipelining
DSP u_FPGA_Core/Gain_mul_temp input u_FPGA_Core/Gain_mul_temp/A[29:0] is not
pipelined. Pipelining DSP48 input will improve performance.
Related violations: <none>
```

```
DPIP-1#2 Warning
Input pipelining
```

DSP u_FPGA_Core/Gain_mul_temp__0 input u_FPGA_Core/Gain_mul_temp__0/A[29:0] is not pipelined. Pipelining DSP48 input will improve performance.
Related violations: <none>

DPOP-1#1 Warning

PREG Output pipelining

DSP u_FPGA_Core/Gain_mul_temp output u_FPGA_Core/Gain_mul_temp/P[47:0] is not pipelined (PREG=0). Pipelining the DSP48 output will improve performance and often saves power so it is suggested whenever possible to fully pipeline this function. If this DSP48 function was inferred, it is suggested to describe an additional register stage after this function. If the DSP48 was instantiated in the design, it is suggested to set the PREG attribute to 1.

Related violations: <none>

DPOP-1#2 Warning

PREG Output pipelining

DSP u_FPGA_Core/Gain_mul_temp__0 output u_FPGA_Core/Gain_mul_temp__0/P[47:0] is not pipelined (PREG=0). Pipelining the DSP48 output will improve performance and often saves power so it is suggested whenever possible to fully pipeline this function. If this DSP48 function was inferred, it is suggested to describe an additional register stage after this function. If the DSP48 was instantiated in the design, it is suggested to set the PREG attribute to 1.

Related violations: <none>

DPOP-2#1 Warning

MREG Output pipelining

DSP u_FPGA_Core/Gain_mul_temp multiplier stage u_FPGA_Core/Gain_mul_temp/P[47:0] is not pipelined (MREG=0). Pipelining the multiplier function will improve performance and will save significant power so it is suggested whenever possible to fully pipeline this function. If this multiplier was inferred, it is suggested to describe an additional register stage after this function. If there is no registered adder/accumulator following the multiply function, two pipeline stages are suggested to allow both the MREG and PREG registers to be used. If the DSP48 was instantiated in the design, it is suggested to set both the MREG and PREG attributes to 1 when performing multiply functions.

Related violations: <none>

DPOP-2#2 Warning

MREG Output pipelining

DSP u_FPGA_Core/Gain_mul_temp__0 multiplier stage u_FPGA_Core/Gain_mul_temp__0/P[47:0] is not pipelined (MREG=0). Pipelining the multiplier function will improve performance and will save significant power so it is suggested whenever possible to fully pipeline this function. If this multiplier was inferred, it is suggested to describe an additional register stage after this function. If there is no registered adder/accumulator following the multiply function, two pipeline stages are suggested to allow both the MREG and PREG registers to be used. If the DSP48 was instantiated in the design, it is suggested to set both the MREG and PREG attributes to 1 when performing multiply functions.

Related violations: <none>

